PYTHON FOR DATA SCIENCE · LESSON 14 OF 31

Debugging, Testing, Regular Expressions and Code Quality

Find and fix bugs systematically, write automated tests with pytest, clean text with regular expressions, and keep code readable with PEP 8, type hints, linters and formatters.

Python class 10 | Python 3.13 · pytest 9 · ruff | Intermediate | 100 min

By the end of this lesson you can

- Read a traceback and debug with print, breakpoints and pdb
- Write and run unit tests with assert and pytest, including parametrised tests
- Use regular expressions to search, extract, validate and replace text
- Apply PEP 8 naming and layout, type hints, and automatic tools (ruff, black, mypy)
- Use AI assistants to debug and review code responsibly

Reading a traceback

Read a traceback **from the bottom up**: the last line names the error, the lines above show where it happened.

Python

```
def average(values):
    return sum(values) / len(values)
def report(data):
    return f"Average: {average(data['scores']):.1f}"
print(report({"scores": []}))
```

OUTPUT

```
ZeroDivisionError: division by zero
```

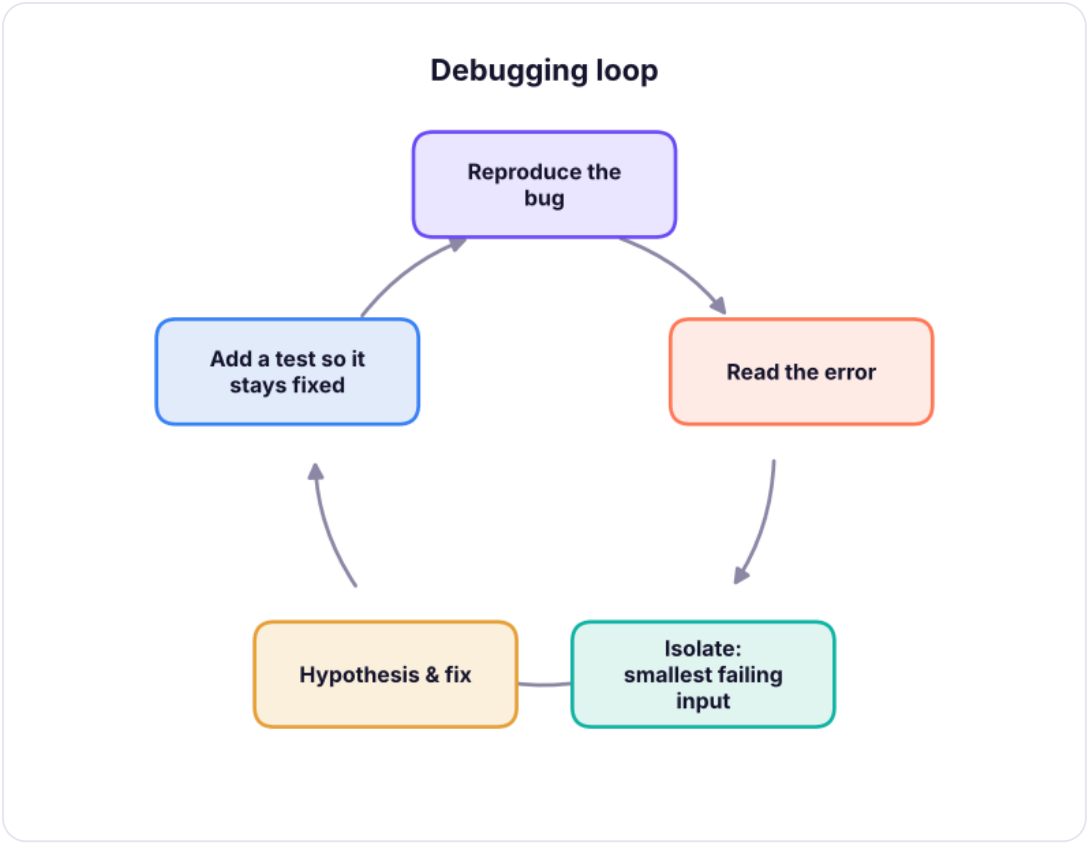
The last line says `ZeroDivisionError`; the frame above points at `average`, called from `report`. The fix is a guard for empty input:

```python
def average(values):
    return sum(values) / len(values) if values else None
print(average([]), average([4, 6]))
```

OUTPUT

```
None 5.0
```

A debugging routine



A systematic debugging loop beats random changes.

Technique	How
Print / f-string debugging	`print(f"{i=} {total=}")` shows names and values
`breakpoint()`	Pauses and opens the `pdb` debugger at that line
IDE debugger	VS Code / PyCharm: breakpoints, step over/into, watch variables
Jupyter `%debug`	Post-mortem debugging after an error in a cell
Rubber-duck	Explain the code line by line to someone (or an AI)
Bisect	Comment out halves of the code to find the failing part

Python

```python
total = 0
for i, v in enumerate([5, 10, 15]):
    total += v
    print(f"{i=} {v=} {total=}")          # the = specifier prints the name and the value
```

OUTPUT

```
i=0 v=5 total=5
i=1 v=10 total=15
i=2 v=15 total=30
```

pdb commands: `n` (next line), `s` (step into), `c` (continue), `p var` (print), `l` (list code), `q` (quit).

Testing

A **test** is code that checks other code. Tests catch regressions when you change things later.

Python

```python
def clean_city(raw):
    return " ".join(raw.split()).title() if raw else None

assert clean_city("  new   delhi ") == "New Delhi"
assert clean_city("") is None
print("asserts passed")
```

OUTPUT

```
asserts passed
```

pytest

`pytest` finds files named `test_*.py` and runs functions named `test_*`. Plain `assert` statements give detailed failure messages.

Python

```
from pathlib import Path
import subprocess, sys
Path("tests_demo").mkdir(exist_ok=True)
Path("tests_demo/cleaning.py").write_text('''
def clean_city(raw):
    return " ".join(raw.split()).title() if raw else None

def to_inr(value):
    return f"₹{value:,.0f}"
''', encoding="utf-8")
Path("tests_demo/test_cleaning.py").write_text('''
import pytest
from cleaning import clean_city, to_inr

@pytest.mark.parametrize("raw, expected", [("  pune", "Pune"), ("NEW  DELHI", "New Delhi"),
("", None), (None, None)])
def test_clean_city(raw, expected):
    assert clean_city(raw) == expected

def test_to_inr():
    assert to_inr(1234567) == "₹1,234,567"

def test_bad_input():
    with pytest.raises(ValueError):          # a str cannot take the ',' number format
        to_inr("abc")

def test_deliberately_failing():
    assert to_inr(1000) == "₹1000"          # wrong expectation, to show a failure report
''', encoding="utf-8")
r = subprocess.run([sys.executable, "-m", "pytest", "-q", "--no-header", "-p",
"no:cacheprovider", "tests_demo"], capture_output=True, text=True)
lines = r.stdout.strip().splitlines()
print("\n".join([l for l in lines if "assert" in l or "passed" in l or "failed" in l]
[-4:]))
import shutil; shutil.rmtree("tests_demo")
```

OUTPUT

```
>       assert to_inr(1000) == "₹1000"          # wrong expectation, to show a
failure report
E       AssertionError: assert '₹1,000' == '₹1000'
1 failed, 6 passed in 0.02s
```

The report says which test failed and shows both values: `'₹1,000' == '₹1000'`. The function is right, so the test expectation is fixed.

> **TIP** **What to test in data work**
>
> Cleaning functions, metric calculations, edge cases (empty, missing, negative, duplicates), schema checks on loaded data (columns, types, ranges), and that a pipeline's row counts reconcile. Tools such as **pandera** and **Great Expectations** validate DataFrames.

Regular expressions

A **regular expression** (regex) is a pattern for text. Python's `re` module searches, extracts, validates and replaces.

Pattern	Matches
`\d` / `\D`	A digit / a non-digit
`\w` / `\s`	Word character (letter, digit, _) / whitespace
`.`	Any character except newline
`[A-Z]`, `[^0-9]`	Character set / negated set
`*`, `+`, `?`	0 or more, 1 or more, 0 or 1
`{3}`, `{2,4}`	Exactly 3, between 2 and 4
`^`, `$`	Start, end of string
`( )`	Capture group; `(?P<name> )` named group
`|`	Or

Python

```
import re
text = "Call 98765-43210 or +91 91234 56789; email asha.k@gmail.com, ravi@skillnavora.in.
Order #A-1029 on 2025-04-07."
print(re.findall(r"\d{5}[- ]?\d{5}", text))                              # phone numbers
print(re.findall(r"[\w.+-]+@[\w-]+(?:\.[\w-]+)+", text))                  # emails (no
trailing full stop)
m = re.search(r"(?P<y>\d{4})-(?P<m>\d{2})-(?P<d>\d{2})", text)
print(m.group(), m["y"], m.groupdict())
print(re.sub(r"\d", "X", "PAN ABCDE1234F"), re.split(r"[;,]\s*", "a, b;c ,d"))
print(bool(re.fullmatch(r"[A-Z]{5}\d{4}[A-Z]", "ABCDE1234F")), bool(re.fullmatch(r"[A-Z]
{5}\d{4}[A-Z]", "ABCD1234F")))    # PAN format
```

OUTPUT

```
['98765-43210', '91234 56789']
['asha.k@gmail.com', 'ravi@skillnavora.in']
2025-04-07 2025 {'y': '2025', 'm': '04', 'd': '07'}
PAN ABCDEXXXXF ['a', 'b', 'c ', 'd']
True False
```

Always use **raw strings** `r"..."` for patterns, so `\d` is not treated as an escape. Test patterns at regex101.com.

Regex on a DataFrame

Python

```python
import pandas as pd
c = pd.DataFrame({"name": ["Asha", "Ravi", "Meera", "Kabir", "Zoya"],
                  "email": ["asha@gmail.com", "ravi@company", "MEERA@Yahoo.co.in", "kabir
@gmail.com", None],
                  "phone": ["98765 43210", "+91-91234-56789", "12345", "9123456789",
"0987654321"]})
c["email_ok"] = c["email"].str.fullmatch(r"[\w.+-]+@[\w-]+(?:\.[\w-]+)+", na=False)
c["domain"] = c["email"].str.lower().str.extract(r"@([\w.-]+)$")[0]
digits = c["phone"].str.replace(r"\D", "", regex=True).str[-10:]
c["phone_ok"] = digits.str.fullmatch(r"[6-9]\d{9}")
print(c[["name", "email_ok", "domain", "phone_ok"]])
```

OUTPUT

```
    name  email_ok       domain  phone_ok
0   Asha      True    gmail.com      True
1   Ravi     False      company      True
2  Meera      True  yahoo.co.in     False
3  Kabir     False    gmail.com      True
4   Zoya     False          NaN     False
```

pandas string methods (`.str.contains` , `.str.extract` , `.str.replace` , `.str.fullmatch`) accept regex and run on the whole column.

Code quality

PEP 8 is Python's style guide. The essentials:

Rule	Example
`snake_case` for variables and functions	`total_sales`, `clean_city()`
`PascalCase` for classes	`CleaningPipeline`
`UPPER_CASE` for constants	`GST_RATE = 0.18`
4-space indentation, lines ≤ 79–100 chars	
Spaces around operators, after commas	`x = a + b`, `f(a, b)`
Imports at the top, grouped std / third-party / local	
Meaningful names over comments	`days_since_last_order`, not `d`

Let tools do the work:

```text
pip install ruff mypy
ruff check .          # lint: unused imports, undefined names, bad patterns
ruff format .         # auto-format (black-compatible)
mypy src/             # check type hints
pre-commit install    # run them automatically before each commit
```

> **NOTE** **Using AI to debug**
>
> Paste the **full traceback and the smallest code that reproduces it** into an AI assistant (Claude, ChatGPT, Copilot, Gemini). Ask it to explain the cause before proposing a fix, then run the fix and add a test. Never paste secrets or confidential data, and check that suggested library functions really exist.

Summary

KEY TAKEAWAYS

- Read tracebacks bottom-up; reproduce, isolate, hypothesise, fix, then add a test.
- Debug with `print(f"{x=}")`, `breakpoint()`, an IDE debugger or `%debug`.
- `pytest` runs `test_*` functions with plain `assert`; use `parametrize` for many cases and `pytest.raises` for errors.
- Regex: `re.search`, `findall`, `fullmatch`, `sub`, `split`; raw strings; pandas `.str` methods accept regex.
- Follow PEP 8 and automate it with ruff, a formatter and mypy.

Quick quiz

6 questions · tap an answer to check it

1. Where do you start reading a Python traceback?

A The first line

B The last line, which names the error

C The middle

D The import lines

Answer: B. Then read upward to find where it happened.

2. What does `print(f"{total=}")` print when total is 5?

A 5

B total=5

C {total=5}

D An error

Answer: B. The `=` specifier is a debugging shortcut.

3. Which file name will pytest discover by default?

A cleaning_tests.py

B test_cleaning.py

C tests.txt

D cleaning.py

Answer: B. Files `test_*.py` or `*_test.py`, functions `test_*`.

4. What does `\d{3}` match?

A The letter d three times

B Exactly three digits

C Up to three digits

D Any three characters

Answer: B. `\d` is a digit; `{3}` repeats it exactly three times.

5. Why use `r"..."` for regex patterns?

A It runs faster

B Backslashes are kept literally instead of being treated as escapes

C It makes the pattern case-insensitive

D It is required by pandas

Answer: B. Raw strings avoid double escaping.

6. Which naming follows PEP 8 for a function?

A CalculateMargin

B calculate_margin

C CALCULATE_MARGIN

D calculateMargin

Answer: B. snake_case for functions and variables.

EXERCISE **Practice**

(1) Write a regex that extracts all hashtags from a tweet. (2) Validate Indian PIN codes (6 digits, not starting with 0). (3) Write three pytest tests for a `percent_change` function, including the zero case. (4) Run `ruff check` on one of your notebooks exported as `.py` and fix the warnings.

Show solution (worked answer)

Python

```python
import re
print(re.findall(r"#\w+", "Loving #Python and #DataScience!"))
pin = re.compile(r"[1-9]\d{5}")
print(bool(pin.fullmatch("411001")), bool(pin.fullmatch("011001")))
# test_metrics.py
# def test_up(): assert percent_change(100, 120) == 20
# def test_down(): assert percent_change(100, 80) == -20
# def test_zero(): assert percent_change(0, 5) is None
```

INTERVIEW CORNER **Questions you may be asked**

Q: How do you debug a function that gives wrong output only for some inputs? Reproduce with the smallest failing input, inspect intermediate values (prints or debugger), form a hypothesis, fix, and add the case as a regression test.

Q: Unit test vs integration test? A unit test checks one small function in isolation; an integration test checks that components work together (e.g. load → transform → write).

Q: Write a regex for a 10-digit Indian mobile number. `^[6-9]\d{9}$` (optionally allowing a `+91` prefix: `^(?:\+91[- ]?)?[6-9]\d{9}$`).

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